

Module 10: Multiple Granularity Locking and Intention Locks

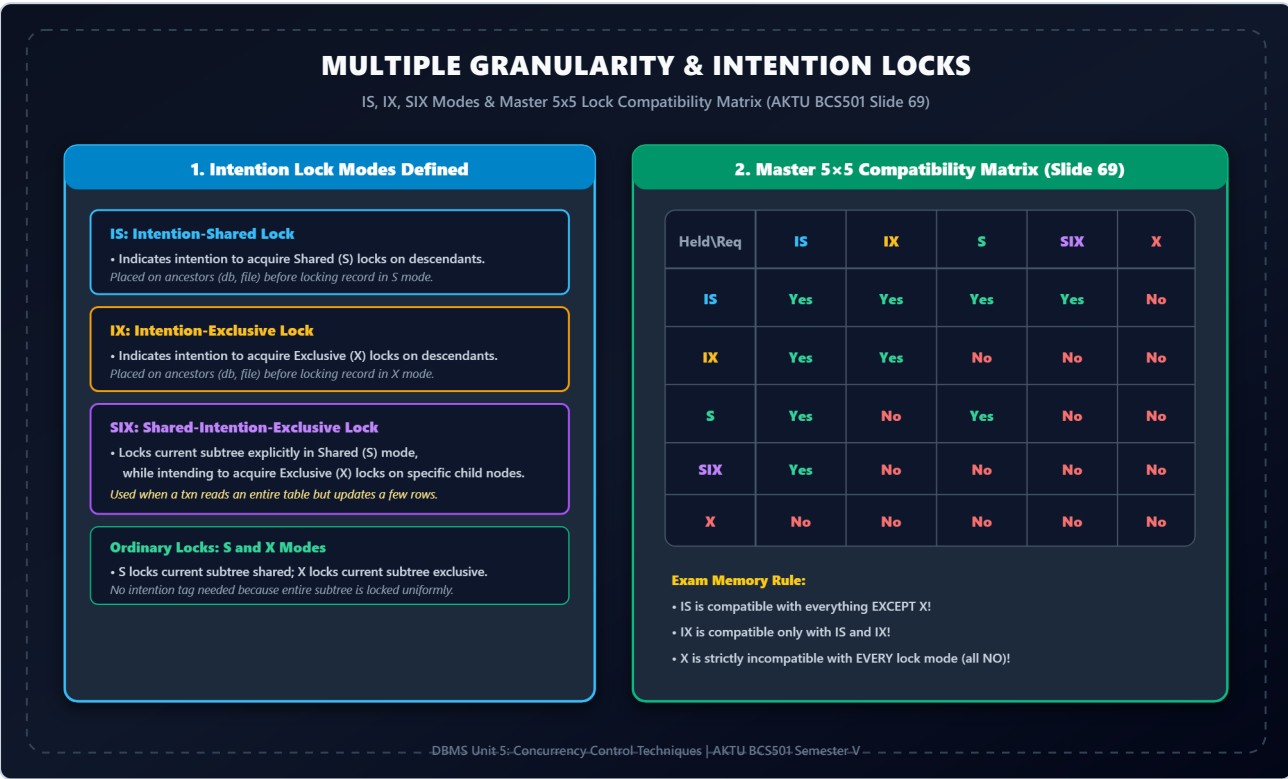


Figure 10.1: Intention Locks (IS, IX, SIX) & Master 5x5 Compatibility Matrix

1. Purpose of Intention Locks

Intention locks avoid full ancestor-descendant tree traversal when checking lock conflicts. Before locking a leaf in S or X mode, intention locks (IS or IX) are placed on all ancestor nodes.

2. The Master 5x5 Lock Compatibility Matrix

Held \ Requested	IS	IX	S	SIX	X
IS	Yes	Yes	Yes	Yes	No
IX	Yes	Yes	No	No	No
S	Yes	No	Yes	No	No
SIX	Yes	No	No	No	No
X	No	No	No	No	No